

Calc 1 – Limits

Worksheet A

1. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.
2. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$.
3. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 4}{2x^2 + 7}$.
4. Use the squeeze theorem to find $\lim_{x \rightarrow 0} x^2 \cos(1/x)$.
5. Use L'Hôpital to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

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Worksheet B

1. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6}$.
2. Evaluate $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$.
3. Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$.
4. Determine $\lim_{x \rightarrow 0^+} x \ln x$ (rewrite, then L'Hôpital).
5. Evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. Factor: $\frac{(x-3)(x+3)}{x-3} = x+3 \rightarrow \boxed{6}$.
2. $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} \rightarrow 5 \cdot 1 = \boxed{5}$.
3. Divide top/bottom by x^2 : $\frac{3-1/x+4/x^2}{2+7/x^2} \rightarrow \boxed{3/2}$.
4. Since $-x^2 \leq x^2 \cos(1/x) \leq x^2$ and both bounds $\rightarrow 0$, limit = $\boxed{0}$.
5. Form 0/0. LH: $\frac{e^x-1}{2x}$, still 0/0. LH again: $\frac{e^x}{2} \rightarrow \boxed{1/2}$.

Worksheet B.

1. Factor: $\frac{(x-2)(x+2)}{(x-2)(x-3)} = \frac{x+2}{x-3} \rightarrow \frac{4}{-1} = \boxed{-4}$.
2. As $x \rightarrow 0$, $\tan 3x \sim 3x$ and $\sin 2x \sim 2x$, so ratio $\rightarrow \boxed{3/2}$.
3. Rationalize: $\frac{(\sqrt{x^2+x}-x)(\sqrt{x^2+x}+x)}{\sqrt{x^2+x}+x} = \frac{x}{\sqrt{x^2+x}+x} \rightarrow \frac{1}{1+1} = \boxed{1/2}$.
4. Rewrite as $\frac{\ln x}{1/x}$ (form $-\infty/\infty$). LH: $\frac{1/x}{-1/x^2} = -x \rightarrow \boxed{0}$.
5. $\ln L = \lim x \ln(1 + 2/x) = \lim x \cdot \frac{2}{x} = 2$, so $L = \boxed{e^2}$.